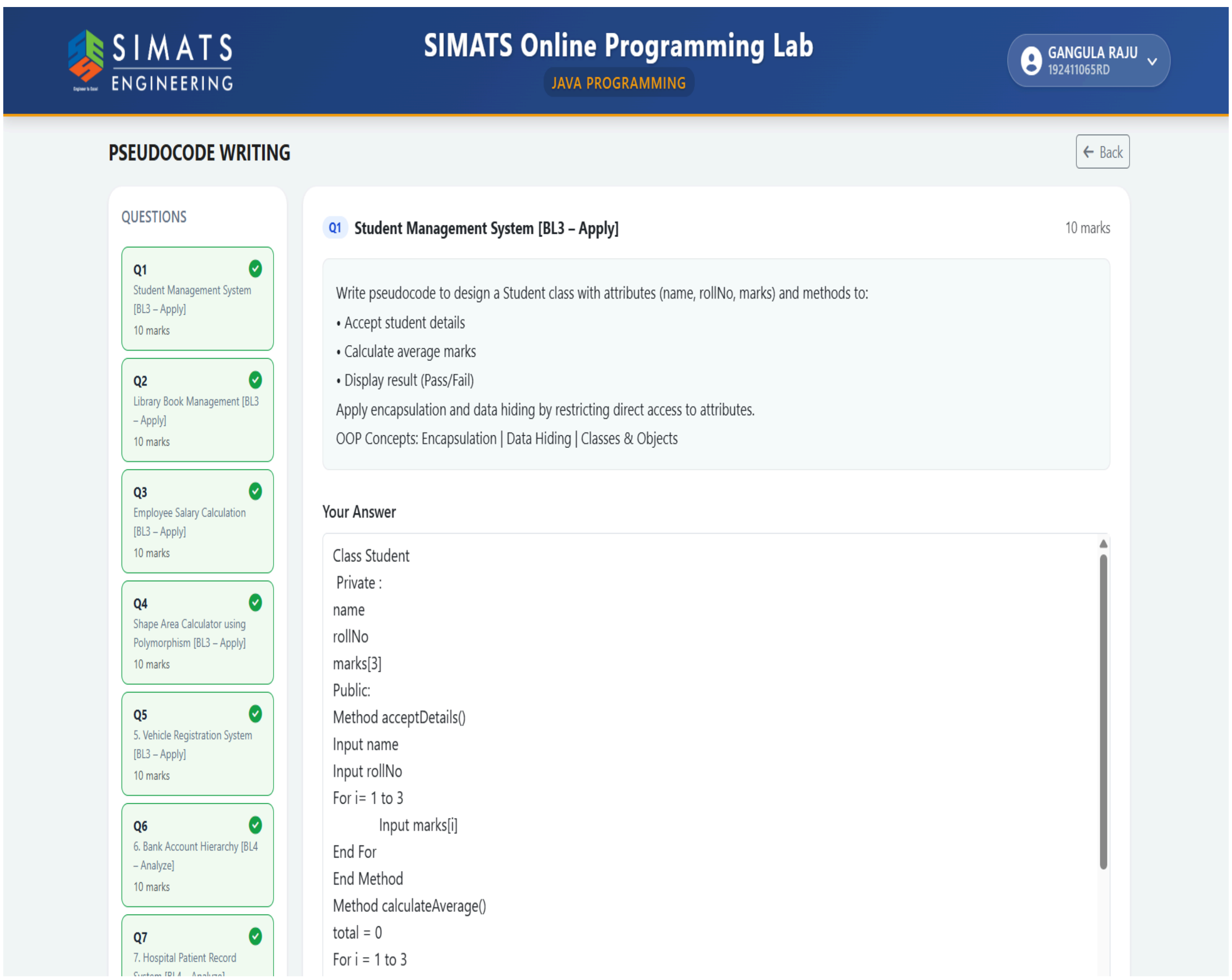
CSA0927 - PROGRAMMING IN JAVA

NAME : G.RAJU
REG NO : 192411065
FACULTY : S.PUSPHALATHA

Q1) STUDENT MANAGEMENT SYSTEM



Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Apply encapsulation and data hiding by restricting direct access to attributes.

OOP Concepts: Encapsulation | Data Hiding | Classes & Objects

Your Answer

```
End For
average = total/3
Return average
End Method
Method displayResult()
avg = calculateAverage()
Print "Name:",name
Print"Roll No :",avg
If avg >=40 Then
Print "Result:Pass"
Else
Print"Result:PASS"
Else
Print "Result : FAIL"
End If
End Method
End class
Main program
Create object S of student
S.acceptDetails()
S.displayResult()
End Main
```

Save Answer

Q2) LIBRARY BOOK MANAGEMENT

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QUESTIONS

Q1
Student Management System [BL3 – Apply]
10 marks

Q2
Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8

Q2 **Library Book Management [BL3 – Apply]** 10 marks

Design a Book class with private attributes (title, author, ISBN, isAvailable). Implement methods to:

- Issue a book (change availability status)
- Return a book
- Display book details using getter methods

Demonstrate object creation for at least 3 books and simulate issue/return operations.

OOP Concepts: Encapsulation | Getters & Setters | Objects

Your Answer

```
Class Book
Private:
title,author,ISBN,isAvailable
Public :
setDetails(t,a,i)
title=t
author=a
ISBN=i
isAvailable=True
issueBook()
isAvailable=False
returnBook()
isAvailable=False
display()
print title,author,ISBN,isAvailable
End class
Main
Create B1,B2,B3
B1.setDetails("Java","James","101")
```

Q1
Student Management System [BL3 – Apply]
10 marks

Q2
Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Design a Book class with private attributes (title, author, ISBN, isAvailable). Implement methods to:

- Issue a book (change availability status)
- Return a book
- Display book details using getter methods

Demonstrate object creation for at least 3 books and simulate issue/return operations.

OOP Concepts: Encapsulation | Getters & Setters | Objects

Your Answer

```
issueBook()
isAvailable=False
returnBook()
isAvailable=False
display()
print title,author,ISBN,isAvailable
End class
Main
Create B1,B2,B3
B1.setDetails("Java","James","101")
B2.setDetails("Python",Guido","102")
B3.setDetails("C++","Bjarne","103")
B1.issueBook()
B1.returnBook()

B1.display()
B2.display()
B3.display()
End Main
```

Save Answer

Q3) EMPLOYEE SALARY CALCULATION

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QUESTIONS

Q1
Student Management System [BL3 – Apply]
10 marks

Q2
Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q3 **Employee Salary Calculation [BL3 – Apply]** 10 marks

Write pseudocode for an Employee class with encapsulated attributes (empId, name, basicPay, designation). Implement methods to: • Calculate gross salary (basic + HRA + DA)

- Calculate net salary after deductions (PF, tax)
- Display pay slip

Apply data hiding so salary components are not directly accessible.

OOP Concepts: Encapsulation | Data Hiding | Methods

Your Answer

```
Class Employee
Private:
empld,name,basicPay,designation
Public:
setDetails(id,n,pay,des)
empld=id
name= n
basicpay=pay
designation=des

calculateSalary()
HRA=basicPay * 20/100
DA = basicpay *10/100
Gross = basicpay+HRA+DA
PF=basicpay *5/100
Tax = gross -(PF +Tax)

display()
```

Q2
Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]

• Display pay slip

Apply data hiding so salary components are not directly accessible.

OOP Concepts: Encapsulation | Data Hiding | Methods

Your Answer

```
calculateSalary()
HRA=basicPay * 20/100
DA = basicpay *10/100
Gross = basicpay+HRA+DA
PF=basicpay *5/100
Tax = gross -(PF +Tax)

display()
   Print empId,name,designation
   Print"Gross S alary =",Gross
   Print "Net Salary=",Net
End Class

Main

Create E
E.setDetails(101,"Raju",30000,"Developer")
E.calculateSalary()
E.display()

End Main
```

Save Answer

Q4) SHAPE AREA CALCULATOR USING POLYMORPHISM

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QUESTIONS

Q1
Student Management System [BL3 – Apply]
10 marks

Q2
Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8

Q4 **Shape Area Calculator using Polymorphism [BL3 – Apply]** 10 marks

Create a Shape base class with an overridable calculateArea() method. Apply this to:

• Circle — area using radius
• Rectangle — area using length and breadth
• Triangle — area using base and height

Use method overriding to demonstrate runtime polymorphism. Call calculateArea() for each object.

OOP Concepts: Inheritance | Polymorphism | Method Overriding

Your Answer

```
Class Shape
  Method calculateArea()
End Class
Class Circle Inherits Shape
Method calculateArea(r)
Print 3.14*r*r
End class
class Rectangle Inherits Shape
Method calculateArea(1,b)
Print 1*b
End class
Class Triangle Inherits Shape
Method calculateArea(base,height)
Print 0.5*base*height
end class
Main
Create Circle C
Create Rectangle R
Create Triangle T
```

Q2
Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]

• Triangle — area using base and height
Use method overriding to demonstrate runtime polymorphism. Call calculateArea() for each object.
OOP Concepts: Inheritance | Polymorphism | Method Overriding

Your Answer

```
Class Shape
  Method calculateArea()
End Class
Class Circle Inherits Shape
Method calculateArea(r)
Print 3.14*r*r
End class
class Rectangle Inherits Shape
Method calculateArea(1,b)
Print 1*b
End class
Class Triangle Inherits Shape
Method calculateArea(base,height)
Print 0.5*base*height
end class
Main
Create Circle C
Create Rectangle R
Create Triangle T
C.calculateArea(5)
C.calculateArea(4,6)
T.calculateArea(8,5)
End Main
```

Save Answer

Q5) VEHICLE REGISTRATION SYSTEM

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QUESTIONS

Q1
Student Management System [BL3 – Apply]
10 marks

Q2
Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q5 **5. Vehicle Registration System [BL3 – Apply]** 10 marks

Design a Vehicle class with attributes (regNo, owner, fuelType). Extend it into:
• TwoWheeler — with engine capacity
• FourWheeler — with seating capacity and AC status
Apply inheritance to reuse common attributes and override a displayDetails() method in each subclass.
OOP Concepts: Inheritance | Method Overriding | Subclass

Your Answer

```
Class Vehicle
regNo,owner,fuelType
Method setDetails(r,o,f)
regNo = r
owner = o
fuelType =f
Method displayDetails()
Print regNo,owner,fueType
End class
Class TwoWheeler Inherits Vehicle
engineCapacity
Method displayDetails()
print regNo,owner,fuelType,engineCapacity
End Class
Class FourWheeler Inherits Vehicle
engineCapacity
Method displayDetails()
print regno,owner,fuel type,seatingcpacity,Ac
End class
```

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

10/10 answered

```
Method setDetails(r,o,f)
regNo = r
owner = o
fuelType =f
Method displayDetails()
Print regNo,owner,fueType
End class
Class TwoWheeler Inherits Vehicle
engineCapacity
Method displayDetails()
print regNo,owner,fuelType,engineCapacity
End Class
Class FourWheeler Inherits Vehicle
engineCapacity
Method displayDetails()
print regno,owner,fuel type,seatingcpacity,Ac
End class
Main
Create T as TwoWheeler
Create F as FourWheeler
T.seetDetails("AP01AB1234","RAJU",petrol")
T.engineCapacity=150
T.displaydetails()
F.setDetails("AP02CD5678","Kiran","Diesel")
F.seatingCapacity = 5
F.AC="yes"
F.displaydetails()
End Main
```

Save Answer

Q6) BANK ACCOUNT HIERARCHY

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QUESTIONS

Q1
Student Management System [BL3 – Apply]
10 marks

Q2
Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q6 **6. Bank Account Hierarchy [BL4 – Analyze]** 10 marks

Develop pseudocode for a BankAccount superclass and two subclasses — SavingsAccount (with interest calculation) and CurrentAccount (with overdraft facility). Analyze and implement:

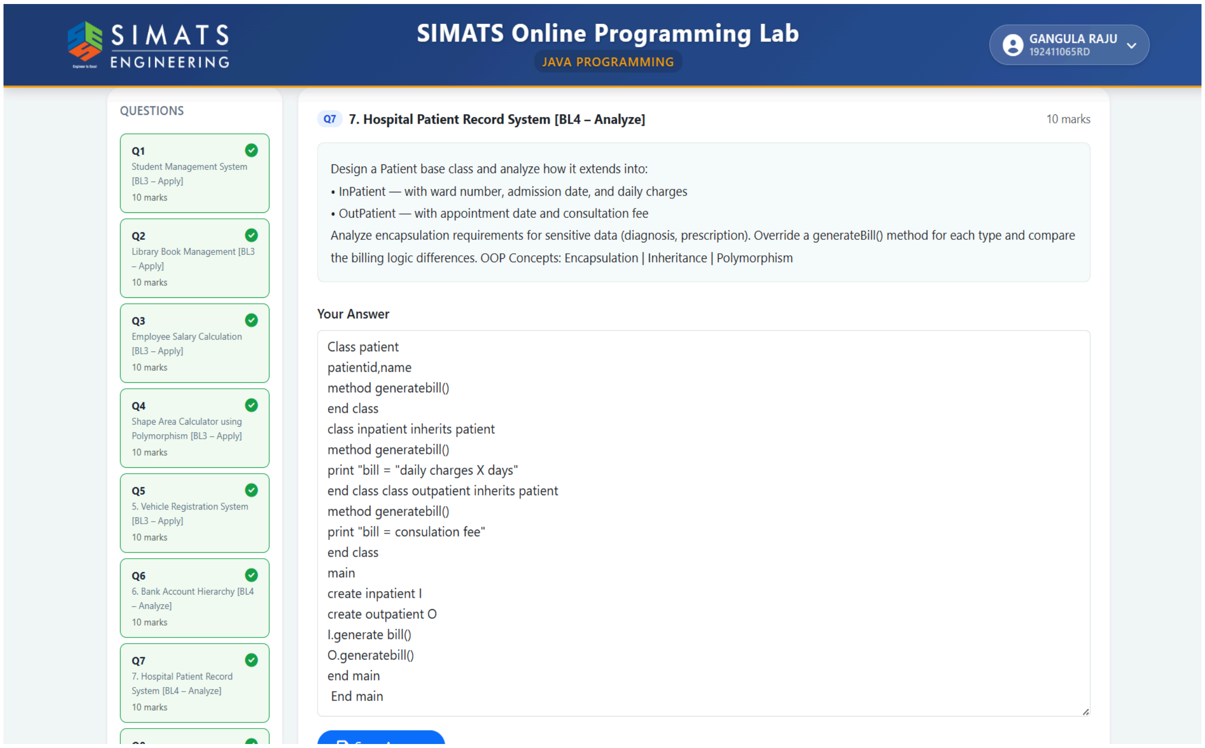
• Inheritance of common attributes (accNo, balance, holder)
• Method overriding for withdrawal behavior
• How overdraft protection differs from savings limit

Compare the behavior of withdraw() across both subclasses and justify the design choices.
OOP Concepts: Inheritance | Method Overriding | Polymorphism

Your Answer

```
Class BankAccont
accno,holder,balance
method withdraw(amount)
end class
class saving account inherits bank account
method withdraw(amount)
print "withdraw with overdraft"
end class
main
create savingaccount s
create current account c
s.withdraw (1000)
c.withdraw(6000)
end main
```

Q7) HOSPITAL PATIENT RECORD SYSTEM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QUESTIONS

Q1
Student Management System [BL3 – Apply]
10 marks

Q2
Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

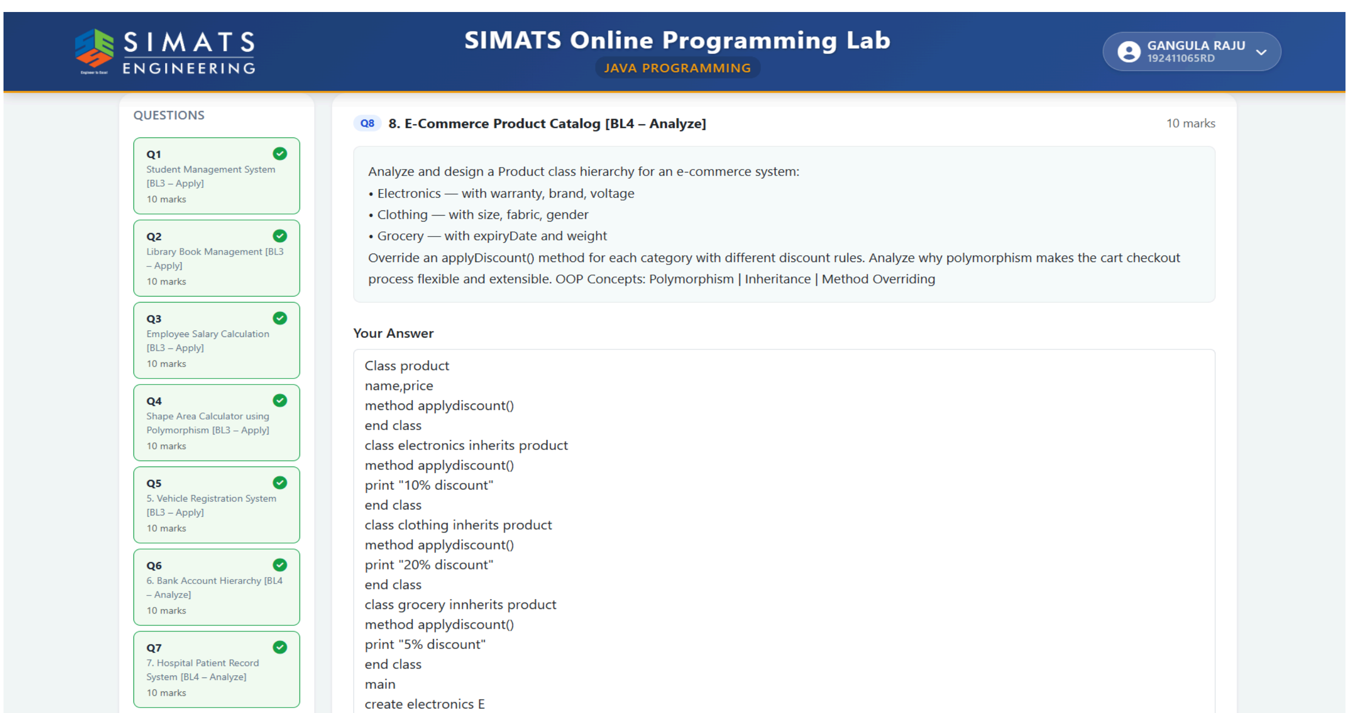
Q7 7. Hospital Patient Record System [BL4 – Analyze] 10 marks

Design a Patient base class and analyze how it extends into:
• InPatient — with ward number, admission date, and daily charges
• OutPatient — with appointment date and consultation fee
Analyze encapsulation requirements for sensitive data (diagnosis, prescription). Override a generateBill() method for each type and compare the billing logic differences. OOP Concepts: Encapsulation | Inheritance | Polymorphism

Your Answer

```
Class patient
patientid,name
method generatebill()
end class
class inpatient inherits patient
method generatebill()
print "bill = "daily charges X days"
end class class outpatient inherits patient
method generatebill()
print "bill = consulation fee"
end class
main
create inpatient I
create outpatient O
I.generate bill()
O.generatebill()
end main
 End main
```

Q8) E-COMMERCE PRODUCT CATALOG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QUESTIONS

Q1
Student Management System [BL3 – Apply]
10 marks

Q2
Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

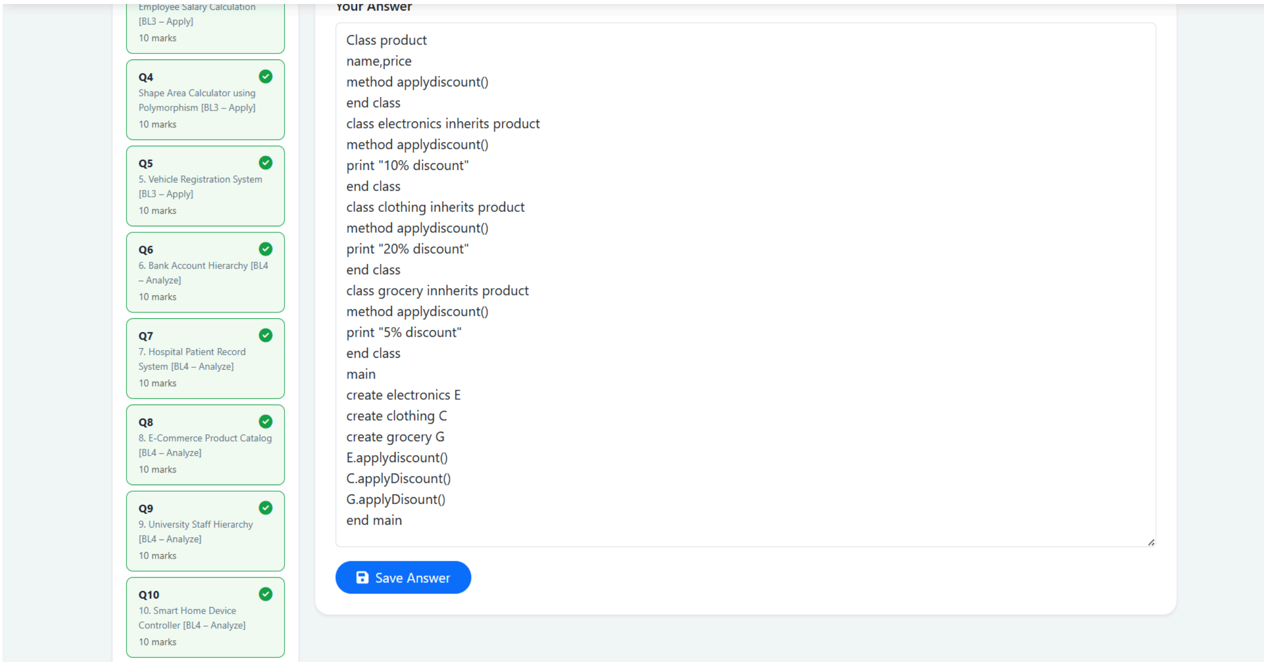
Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8 8. E-Commerce Product Catalog [BL4 – Analyze] 10 marks

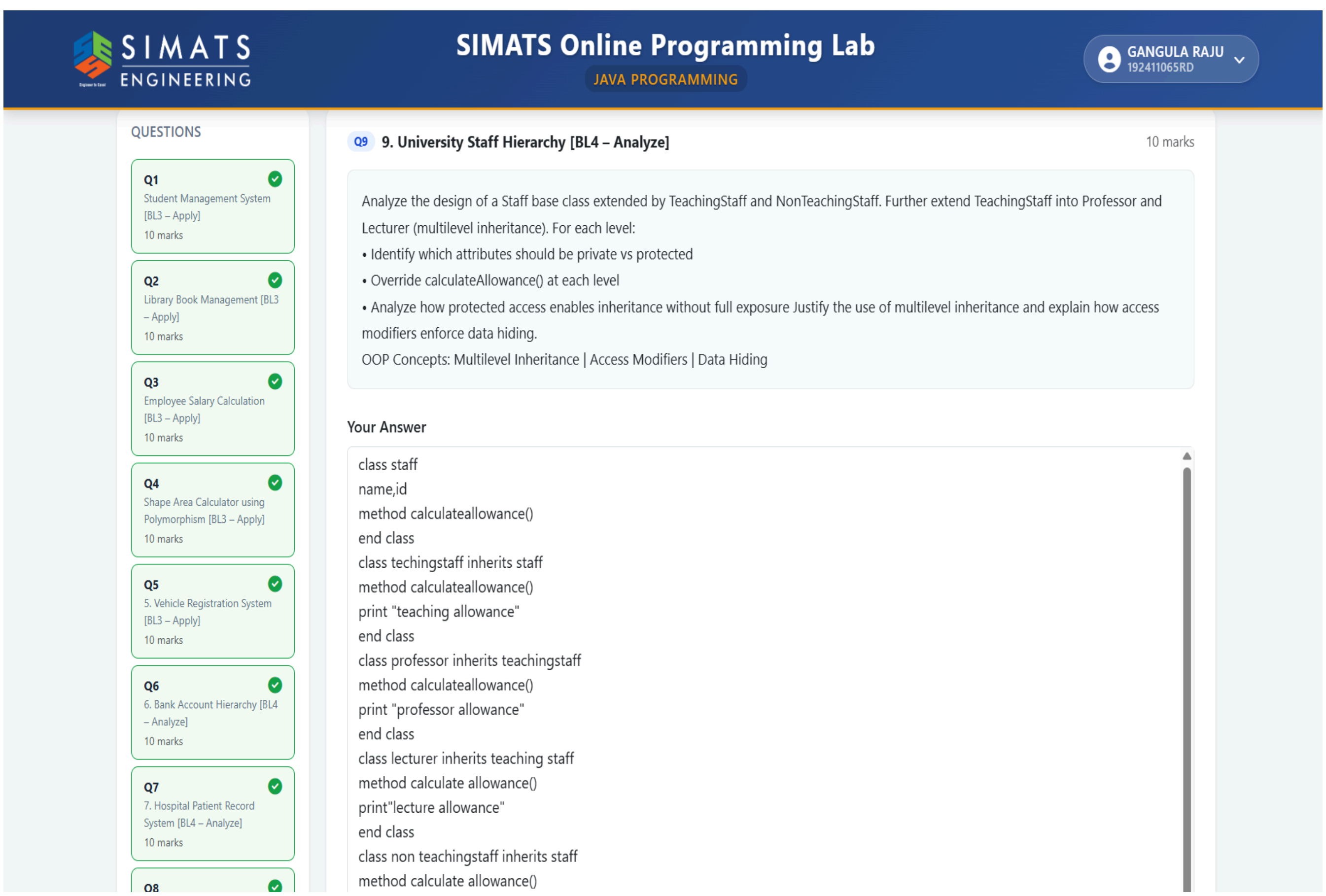
Analyze and design a Product class hierarchy for an e-commerce system:
• Electronics — with warranty, brand, voltage
• Clothing — with size, fabric, gender
• Grocery — with expiryDate and weight
Override an applyDiscount() method for each category with different discount rules. Analyze why polymorphism makes the cart checkout process flexible and extensible. OOP Concepts: Polymorphism | Inheritance | Method Overriding

Your Answer

```
Class product
name,price
method applydiscount()
end class
class electronics inherits product
method applydiscount()
print "10% discount"
end class
class clothing inherits product
method applydiscount()
print "20% discount"
end class
class grocery innherits product
method applydiscount()
print "5% discount"
end class
main
create electronics E
create clothing C
```



Q9) UNIVERSITY STAFF HIERARCHY



Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

10/10 answered

```
class staff
name,id
method calculateallowance()
end class
class techingstaff inherits staff
method calculateallowance()
print "teaching allowance"
end class
class professor inherits teachingstaff
method calculateallowance()
print "professor allowance"
end class
class lecturer inherits teaching staff
method calculate allowance()
print"lecture allowance"
end class
class non teachingstaff inherits staff
method calculate allowance()
print "non-teaching allowance"
end class
create profeesor P
create lecturer L
create non teachingstaff  N
P.calculate allowance()
L.calculate allowance()
N.calcuate allowance()
end main
end main
```

Save Answer

Q10) UNIVERSITY STAFF HIERARCHY

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JAVA PROGRAMMING

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QUESTIONS

Q1
Student Management System [BL3 – Apply]
10 marks

Q2
Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8

Q10 **10. Smart Home Device Controller [BL4 – Analyze]** 10 marks

Analyze and design a SmartDevice superclass with subclasses: SmartLight, SmartThermostat, and SmartLock. Override an executeCommand(cmd) method in each, where the same command (e.g., 'on', 'off', 'status') produces device-specific behavior. Analyze:

- How polymorphism allows a single controller loop to manage all devices
- Security implications of data hiding in SmartLock (PIN must never be exposed) Design the class hierarchy and explain the OOP principles applied at each level.

OOP Concepts: Polymorphism | Encapsulation | Inheritance

Your Answer

```
class smartdevice
method executecommand(cmd)
end class
class smartlight inherits smartdevice
method executecommand(cmd)
end class
class smarthermstat inherits smartdevice
method executeCommand(cmd)
print "light",cmd
end class
class smartthermostat inherits smartdevice
method executecommand(cmd)
print "thermostat :,cmd
end class
class smartlock inherits smartdevice
private PIN
method executecommand(cmd)
private "lock",cmd
```

Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

Your Answer

```
end class
class smartlight inherits smartdevice
method executecommand(cmd)
end class
class smarthermstat inherits smartdevice
method executeCommand(cmd)
print "light",cmd
end class
class smartthermostat inherits smartdevice
method executecommand(cmd)
print "thermostat :,cmd
end class
class smartlock inherits smartdevice
private PIN
method executecommand(cmd)
private "lock",cmd
end class
main
create smartlight L
create smartthermostat T
create smartlock S

L.executecommand("ON")
T.executecommand("STATUS")
S.executecommand("OFF")
End main
```

Save Answer